

IMPERIAL

Imperial College London
Department of Mathematics

Statistics Research Project Title

FORENAME SURNAME

CID: 01234567

Supervised by SUPERVISOR NAME and COSUPERVISOR NAME

24 June 2025

Submitted in partial fulfilment of the requirements for the MSc in Statistics at
Imperial College London

The work contained in this thesis is my own work unless otherwise stated.

Signed: STUDENT'S NAME

Date: DATE

Acknowledgements

In the acknowledgments, you should account for anyone who supported you during your research. This could include family, your research peer group, other students, research group members, your supervisor, data providers, external partners, any computing service used, any financial support.

If your research peer group, other students, research group members, or your supervisors made substantive contributions towards your report – e.g. they identified suitable methods, provided code or contributed a sub-analysis – you must acknowledge their precise contribution here.

Abstract

Abstracts should be no longer than 300 words, and similar in style to a general science or statistics research paper.

General tips. Your research report will be judged on the quality of the work it describes according to the Marking Criteria shared with you. When writing your research report you may assume that the reader is familiar with the four core modules of the MSc in Statistics. Any additional other material should be explained to the reader, providing suitable references. See Section D of the supplementary material for more details. For example, you should avoid writing up in detail statistical methods discussed in one of the four core modules, but instead you should briefly present such methods and add appropriate references. More advanced methods or methods discussed in one of the elective modules should be described in more detail and referenced appropriately.

Your report must be presented within this template and be at most 35 pages in length, as counted by the Arabic page numerals (1,2,3...). This limit includes all figures and tables presented in the main text, excluding the Endmatter and References sections. This is approximately equivalent to a limit of 15,000 words. Shorter submissions may be graded highly, while excess length disproportionate to the content may be penalised.

Before submitting the research project, make sure you read the report in its entirety. There should be no half-finished sentences. All mathematical symbols should be defined. Use a spell-checker.

You will have to submit your report electronically in PDF format through the virtual learning environment (Blackboard). Please name this file in the following format: “CID-Surname-Firstname-Report.pdf”. Your report will be checked for plagiarism via online plagiarism detection services (e.g. Turnitin).

Note the report submission deadline is a very hard deadline since the assessment process has then to be completed on a very short timescale.

1. Introduction

Tips for the Introduction. This is where you describe the topic and the research objectives of your project.

You should attempt to set your work in the context of other work previously done in the field. Convey the background of your research referencing earlier work as appropriate. Define core terms. Include background and context, and focus as appropriate on the wider statistical context. Aim to demonstrate that you can confidently describe your project within a broader context of statistical research, as well as scientific research, that goes well beyond the more narrow topic of your project.

The introduction needs to demonstrate that you are aware of what you are doing, and how it relates to other work that should be properly referenced.

Clearly point out your main contributions at the end of the introduction, and provide an overview how the rest of your report is structured and links together.

Aim for approximately 1.5-3 pages, similar in style to a general science or statistics research paper.¹

2. Methods

Tips for the Methods section. In general, you may structure the main body of your report in a form that suits your project best. We recommend to start with a substantial Methods section that includes, as in a research paper, a description of the following:

- Key definitions and setting using mathematical notation. Any terminology related to the topic/data needs to be clearly presented and explained to the reader.
- Any data used in the project should be clearly explained in sufficient detail in the Methods section so that readers can follow the rest of your report without the need to consult an other report or any of your references.
- If in the project you have generated simulated data, a clear presentation of the generative procedure needs to be included.

¹Tip: If you choose to use footnotes, do so sparingly.

- Background statistical work using mathematical notation.
- All maths should be presented in inline or display formulas with appropriate referencing. For example, use $\exp(x)$ not $\exp(x)$ and $\sin(\theta)$ not $\sin(\theta)$ or $\sin(\theta)$. Display formulas should be numbered using the equation environment if they are referenced in the main text. Display equation blocks should be numbered using the subequations environment:

$$Y = 1 + Z \tag{1a}$$

$$Z \sim \mathcal{N}(0, 1), \tag{1b}$$

so that you can reference (1), (1a), and (1b).

- If your main findings are of an applied nature, you may prefer to describe any new/original statistical procedures or models here. Include the key derivations and proofs if there are any, and any supplementary derivations in the appendix. Clearly highlight your contribution relative to existing work.
- If your main findings are of a theoretical nature, you may prefer to describe any new/original statistical procedures or models in the Results section.
- Structure your Methods section using LaTeX subsections and reference them using labels.
- Aim for approximately 8-15 pages, similar in style to a general science or statistics research paper.

2.1. My Methods subsection

This text is in Subsection 2.1, which is a part of Section 2.

2.1.1. My subsubsection

If needed, subsections can also have subsubsections within them. Keep subsections and subsubsections numbered, which will help us to reference them.

Paragraphs. In the rare event that you need a more deeply nested structure within a subsubsection, the `\paragraph{}` command can be useful.

3. Results

Tips for the Results section.

- If your main findings are of a theoretical nature, you may prefer to describe any new/original statistical procedures or models in the Results section.
- Describe your findings concisely and precisely using scientific language.
- Include sound interpretations and explanations wherever possible to demonstrate your original and clear statistical thought process.
- Use figures and tables to support your main arguments, which should include a caption describing, e.g., what is shown on the x - and y -axis or in the rows and columns. Summarise the main finding or headline numbers in the main text, including references to Figures and Tables as needed. Do not report all numbers shown in a figure or table.
- Include references to previous work as needed.
- Structure your Results section into numbered Subsections and Subsubsections as needed, and feel free to reference these throughout.
- Aim for approximately 5-15 pages, similar in style to a general science or statistics research paper.

4. Discussion

Tips for the Discussion section.

- Begin your discussion with a summary and the main findings of your research in 2-5 sentences.
- Describe the implications of your main findings in the context of existing work concisely and precisely using scientific language.

- Describe the limitations in your statistical methods and your main findings. Be honest about the limitations in your approach, and substantiate what could have been done differently as needed. Explain if your main findings are robust or sensitive to these limitations.
- Avoid Subsections and Subsubsections in the Discussion.
- In the last paragraph, conclude your report with a pitch using plain language that summarises the key implications of your research in the context of previous work. Write this last paragraph for the Imperial Press Office or journalists as audience.
- Aim for approximately 1-3 pages, similar in style to a general science or statistics research paper.

5. Endmatter

Tips for the Endmatter section.

- Include a data availability statement. Examiners must be able to reproduce your analyses on all the data investigated as would be the case during peer review of a research paper.
- Include a reproducibility statement, including the URL under which your code can be found. Examiners must be able to see your code, and they must be able to run your code.
- Include a statement on how large language models or grammar tools were used in the writing of your research report, as a whole. For example, “For this report, GPT-4o was used to copy-edit introduction and conclusion text via ChatGPT graphical user interface. Grammarly was used to improve the quality of written English throughout the report. Where the output of specific queries is included in the document this is clearly quoted and cited as personal communications.”

References

- Alexander, A. (2024). Personal communication - online meeting 2024-08-21.
- Dickens, C. (1859). *A Tale of Two Cities*. Chapman & Hall.
- Efron, B. (1979). Bootstrap Methods: Another Look at the Jackknife. *Annals of Statistics*, 7(1):1–26.
- Efron, B. and Tibshirani, R. J. (1994). *An Introduction to the Bootstrap*. CRC press.
- OpenAI (2025). ChatGPT response: how should I reference this query using BibTeX?
<https://chat.openai.com>. Accessed: 2025-06-24.

A. Supplementary Materials

Tips for the Supplement. Please note that the Supplementary Material is not normally read by your examiners. Include material that you might like to include for your own reference, for reference of others beyond your examiners, or as backup to your thesis in case of further discussion.

Here, we use the Supplement to provide further tips on writing your thesis.

B. Figures

Tips for the Figures. It is better to create figures in a vector-based format, such as PDF. For plots with very many elements these vector-based graphics can become large and slow to load. In these cases a high-resolution bitmap graphic, such as a png, is preferred.

Remember to make fonts in figures large enough to be read once the plots are embedded in the report (compare the two plots within Figure 1). You may need to make multiple versions of the same figure if you plan to use it in your report, poster and presentation.

All figures must be referenced in the main text of the report, in the order that they are referenced.

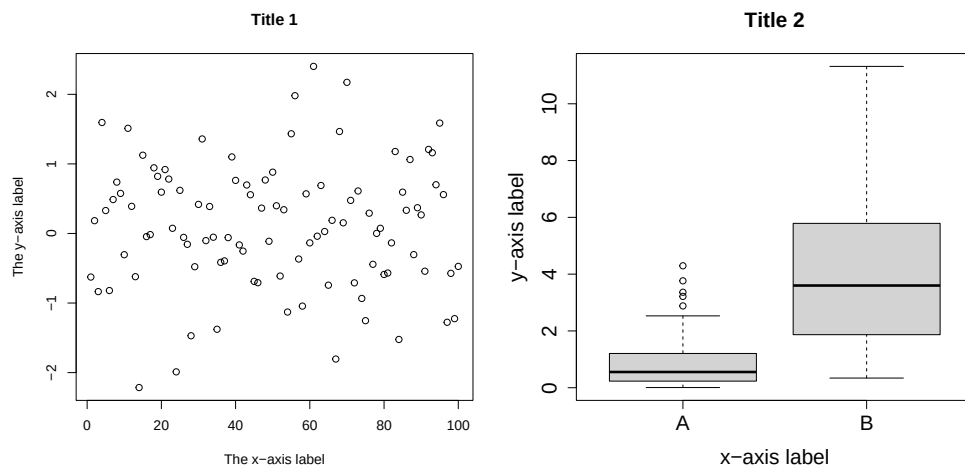


Figure 1.: An example scatterplot (left) and box plot (right).

C. Tables

Tips for the Tables. Tables can be a useful way of displaying summary statistics. An example of a simple table is presented as Table 1. All tables must be referenced in the main text of the report, in the order that they are referenced.

z	$P(Z < z)$
1.281	0.900
1.645	0.950
1.960	0.975
2.326	0.990
2.576	0.995

Table 1.: Partial table showing values of z for $P(Z < z)$, where Z has a standard normal distribution.

D. Referencing sources, sections and items

D.1. Referencing external sources

You should give attribution to the academic books and articles that you draw from and build upon in your research report. You must give full attribution if and how you used generational AI during your research, referencing its use as outlined in our Guidance to using generational AI on the MSc in Statistics.

You should use inline or parenthetical references to your sources. Inline references are used when a source is used as a noun within your sentence. For example, [Efron and Tibshirani \(1994\)](#) describes the bootstrap procedure. Parenthetical references are used to support a claim where the source is not explicitly mentioned within the sentence. For example, the bootstrap procedure stemmed from an earlier approach called the jackknife, which withholds a single data entry at each iteration ([Efron, 1979](#)).

Note that to make the references appear, you will need to compile the bibtex, otherwise you may just see question marks where the references should be. The small selection of example bibtex references used in this template are provided in the file `refs.bib`. These should be replaced with your own bibtex entries.

You can also cite works that cannot be recovered directly by the reader. This is cited as a *personal communication*, which historically would have represented an in-person

conversation or letter exchange. This now extends to many other forms of irrecoverable exchange including: email or instant messaging, interviews, live speeches, meetings and phone calls or notes made during a lecture that were not formally distributed.

You should use a personal communication citation only when a recoverable source is not available and when the communication or work contains original content that is not readily available in any other source. For example, the proprietary models currently used rely strongly on the assumption of independent observations, despite it leading to overly confident parameter estimates ([Alexander, 2024](#)).

To reference the output of a large language model using BibTeX, you can follow a format similar to citing software or personal communication. Here's a recommended format based on academic conventions as of 2025:

```
@misc{chatgpt2025a,  
  author      = {{OpenAI}},  
  title       = {{ChatGPT response to a user query: how should I  
                reference this query using BibTeX?}},  
  year        = {2025},  
  howpublished = {\url{https://chat.openai.com}},  
  note        = {Accessed: 2025-06-24}  
}
```

This can then be referenced within the main text ([OpenAI, 2025](#)).

D.2. Referencing elements of your report

You might also want to cross reference results or claims from another part of your report such as a section, result or equation. This is done by assigning a `\label{}` to the item you wish to cross-reference and then using that label within `\ref{}`. For example, Theorem [E.2](#) is proved in Section [E](#). When referencing displayed mathematics, use `\eqref{}`. As an example, see Equation [\(3\)](#).

D.2.1. Labelling tables and figures

It is important that the label command `\label{LABELNAME}` comes **after** the caption command. See Table 1 and Figure 1 as examples of this.

D.3. Quoting sources

If you wish to quote a source, be sure to use quotation marks and cite the reference. The `\usequote` command is useful here:

“It was the best of times, it was the worst of times, it was the age of wisdom, it was the age of foolishness, it was the epoch of belief, it was the epoch of incredulity, it was the season of Light, it was the season of Darkness, it was the spring of hope, it was the winter of despair, we had everything before us, we had nothing before us, we were all going direct to Heaven, we were all going direct the other way - in short, the period was so far like the present period, that some of its noisiest authorities insisted on its being received, for good or for evil, in the superlative degree of comparison only.” (Dickens, 1859)

E. Definitions, theorems and examples

The following environments are supported: Definition, Theorem, Proof, Proposition, Lemma, Remark, Example.

Definition E.1. The **variance** of a random variable X is defined as

$$\text{Var}(X) = \text{E}[(X - \text{E}[X])^2]. \quad (2)$$

Theorem E.2. Given a random variable X , over all values $a \in \mathbb{R}$,

$$\min_{a \in \mathbb{R}} \text{E}[(X - a)^2] = \text{E}[(X - \text{E}[X])^2]. \quad (3)$$

Proof. Starting with the left-hand side of Equation (3),

$$\begin{aligned}
 \mathbb{E}[(X - a)^2] &= \mathbb{E}[(X - \mathbb{E}[X] + \mathbb{E}[X] - a)^2] \\
 &= \mathbb{E}[(X - \mathbb{E}[X])^2] + 2\mathbb{E}[(X - \mathbb{E}[X])(\mathbb{E}[X] - a)] + \mathbb{E}[(\mathbb{E}[X] - a)^2] \\
 &= \mathbb{E}[(X - \mathbb{E}[X])^2] + (\mathbb{E}[X] - a)^2 \\
 &\geq \mathbb{E}[(X - \mathbb{E}[X])^2],
 \end{aligned}$$

since $\mathbb{E}[X]$ is a real number and $(\mathbb{E}[X] - a)^2 \geq 0$, and the third line follows from linearity of expectation:

$$\mathbb{E}[(X - \mathbb{E}[X])(\mathbb{E}[X] - a)] = (\mathbb{E}[X] - a) \mathbb{E}[(X - \mathbb{E}[X])] = (\mathbb{E}[X] - a) (\mathbb{E}[X] - \mathbb{E}[X]) = 0,$$

since $\mathbb{E}[\mathbb{E}[X]] = \mathbb{E}[X]$, which proves the result. $\square$

Remark E.3. This theorem shows that the minimum of the quantity $\mathbb{E}[(X - a)^2]$ is equal to $\text{Var}(X)$. In some sense, this makes the variance a natural measure of dispersion if we are taking the metric to be the squared deviation of X .

Lemma E.4 (Stein’s Lemma). Let $X \sim N(\mu, \sigma^2)$, and let g be a differentiable function satisfying $\mathbb{E}[|g'(X)|] < \infty$. Then

$$\mathbb{E}[g(X)(X - \mu)] = \sigma^2 \mathbb{E}[g'(X)].$$

Proposition E.5 (Popoviciu’s inequality). Suppose that the random variable X is known to only take values in the bounded range $[a, b]$. Then

$$\text{Var}[X] \leq \frac{(b - a)^2}{4}.$$

Example E.6. Suppose $X \sim \text{Bern}(p)$, for some $p \in [0, 1]$. Then, since $X \in \{0, 1\}$, X is bounded between 0 and 1 and so $\text{Var}[X] \leq \frac{1}{4}$.

Note that Equations (2) and (3) are numbered because they are referred to elsewhere in the main text. All other equations are unnumbered.

Each of these environments may be labelled and referenced as with sections. For example, we might discuss Definition E.1 or Theorem E.2 in this paragraph.